Assignment No 6: Object detection using Transfer Learning of CNN architectures

a. Load in a pre-trained CNN model trained on a large dataset

b. Freeze parameters (weights) in model's lower convolutional layers

c. Add custom classifier with several layers of trainable parameters to model

d. Train classifier layers on training data available for task

e. Fine-tune hyper parameters and unfreeze more layers as needed

In [1]:
```
from tensorflow.keras.preprocessing.image import ImageDataGenerator
```

In [2]:
```
train_dir = 'dataset/mnist-jpg/mnist-jpg/train/'
test_dir = 'dataset/mnist-jpg/mnist-jpg/test/'
```

In [13]:
```
img_gen = ImageDataGenerator(rescale=1.0/255)

data_gen = img_gen.flow_from_directory(
    train_dir,
    target_size=(32,32),
    batch_size=5000,
    shuffle=True,
    class_mode='categorical'
)
```

```
Found 60000 images belonging to 10 classes.
Found 60000 images belonging to 10 classes.
```

In [14]:
```
x_train, y_train = data_gen[0]
x_test, y_test = data_gen[2]
```

In [15]:
```
from tensorflow.keras.applications import VGG16
path = 'dataset/vgg16_weights_tf_dim_ordering_tf_kernels_notop.h5'

vgg_model = VGG16(weights=path,include_top=False, input_shape=(32,32,3))
```

In [16]:
```
for layer in vgg_model.layers:
    layer.trainabler=False
```

In [17]:
```
from tensorflow import keras
from tensorflow.keras.layers import Dense, Flatten, Dropout
```

In [22]:
```
custom_classifier = keras.Sequential([
    Flatten(input_shape=(1,1,512)),
    Dense(100, activation='relu'),
    Dropout(0.2),
    Dense(100, activation='relu'),
    Dropout(0.2),
    Dense(10, activation='softmax')

])

model = keras.Sequential([
    vgg_model,
    custom_classifier
])
```

In [23]:
```
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
```

In [25]:
```
model.fit(x_train, y_train, batch_size=100, epochs=1, validation_data=(x_test,y_test))
```

```
50/50 [==============================] - 204s 4s/step - loss: 1.6244 - accuracy: 0.3820 - val_loss: 1.2011 - val
_accuracy: 0.5770
```

Out[25]: `<keras.src.callbacks.History at 0x22c7ef89c60>`

In [26]:
```
for layer in vgg_model.layers[:-4]:
    layer.trainable = True

model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
model.fit(x_train, y_train, batch_size=1000, epochs=1, validation_data=(x_test,y_test))
```

```
5/5 [==============================] - 208s 42s/step - loss: 2.5760 - accuracy: 0.3306 - val_loss: 2.1499 - val_
accuracy: 0.2630
```

Out[26]: `<keras.src.callbacks.History at 0x22c7822e950>`

In [27]:
```
loss, acc = model.evaluate(x_test, y_test)
```

```
print(loss, " ", acc)
```

```
32/32 [==============================] - 4s 120ms/step - loss: 2.1499 - accuracy: 0.2630
2.1498725414276123   0.2630000114440918
```

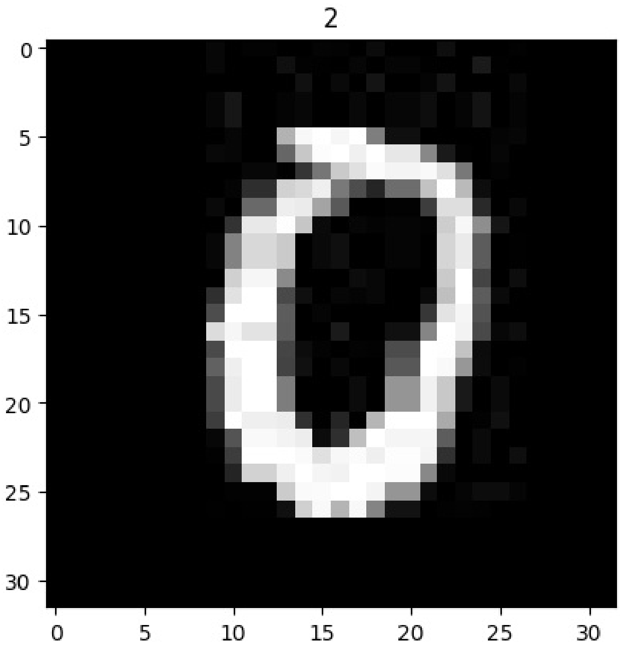
In [28]:
```
pred = model.predict(x_test)
```

```
32/32 [==============================] - 4s 115ms/step
```

In [29]:
```
labels = list(data_gen.class_indices.keys())
```

In [32]:
```
import matplotlib.pyplot as plt
import numpy as np
plt.imshow(x_test[10])
plt.title(str(labels[np.argmax(pred[10])]))
print(str(labels[np.argmax(y_test[10])]))
```

```
0
```



In [33]:
```
y_test[10]
```

Out[33]:
```
array([1., 0., 0., 0., 0., 0., 0., 0., 0., 0.], dtype=float32)
```